

Big group project assignment 2

Team Members: GEOSCIENCE GROUP 3

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Team member 1: [Nithya Reddy Billa](#)

PROJECT CONTEXT:

Dataset: Redlair Weather Station Data (2021–2024)

Station Used: SR Station (Hilltop, Topography Class = 1)

Modules: M1 (Atmospheric sensors), M2/W2M2 (Moisture/well sensors)

Data Types: Air temperature, humidity, solar radiation, dew point, soil temperature, wind speed

Goal: Understand the influence of microtopography (hilltop vs. other stations) on climate behavior.

Redlair weather stations collect environmental data at different microtopographic positions.

Topography classes:

- 1 = Hilltop
- 2 = Mid-Slope
- 3 = Valley / Downstream

Focus Topic:

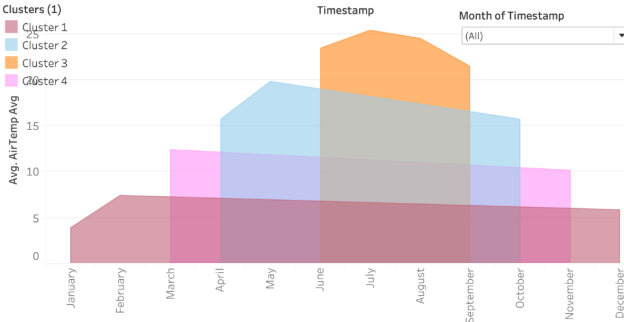
Seasonal Climate Patterns at the SR Hilltop Station (2021–2024)

Lead Student: Nithya Reddy Billa

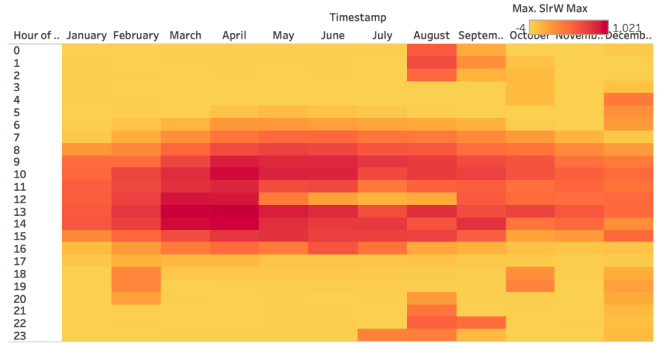
Design A

Hilltop Microclimate Analysis: Multi-Year Temperature, Sunlight, and Moisture Patterns

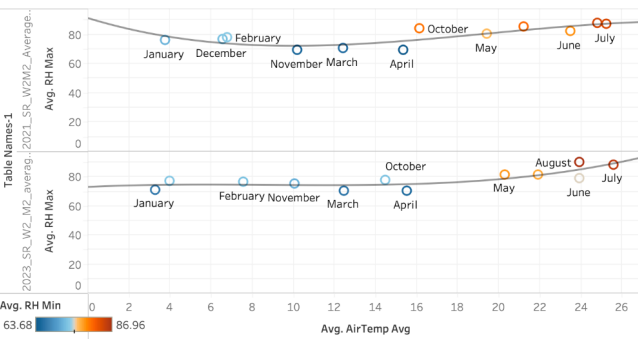
Monthly Air Temperature Patterns at the SR Hilltop Station (2021–2024)



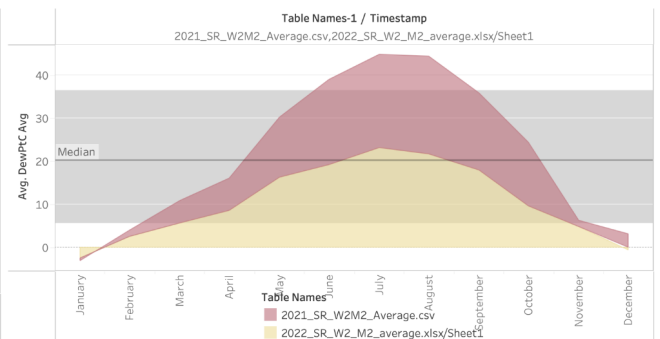
Hourly Sunlight Intensity Across the Year at the hilltop



Temperature–Humidity Interaction Across Months (SR Hilltop)



Hilltop Microclimate Moisture Trend: Seasonal Dew Point Patterns



DISCUSSION:

Design A provides a broad, atmospheric-level view of the hilltop microclimate by combining four key variables: temperature, sunlight, humidity, and dew point. The monthly air temperature chart establishes the overall seasonal cycle, showing how the hilltop warms and cools across the year. The hourly sunlight heatmap adds context by revealing how radiation intensity shifts by month and hour. The temperature–humidity scatterplot shows the relationship between warming temperatures and declining humidity, a signature of elevated, exposed microtopography. Finally, the seasonal dew point trend line highlights how moisture availability follows temperature patterns. Together, these views deliver a complete picture of how hilltop atmospheric conditions evolve seasonally.

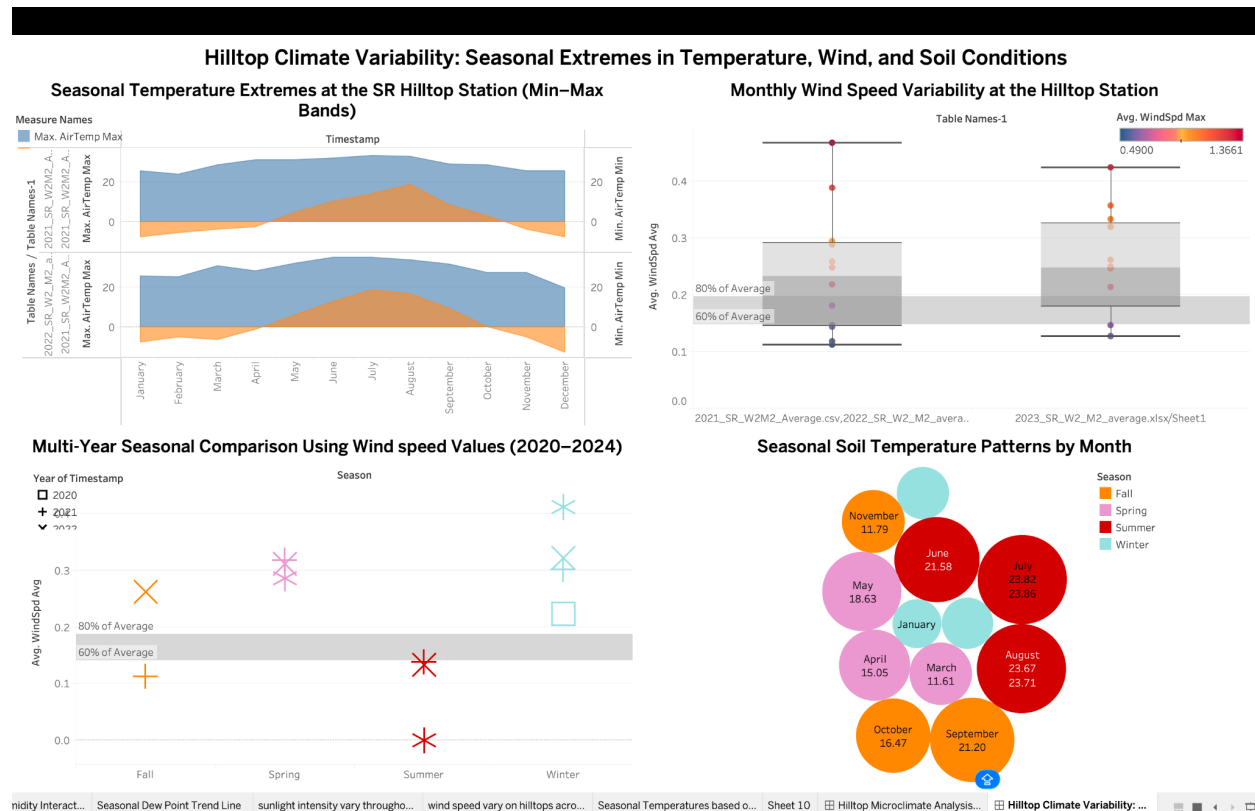
Pros

- Covers the full atmospheric system (temp + sunlight + humidity + moisture)
- Shows clear, intuitive seasonal patterns
- Strong for understanding broad climate behavior at the hilltop
- Easy foundation for topographic comparison with other stations

Cons

- Less emphasis on extremes or high-variability events
- Lacks wind speed analysis
- Dew-point trend may require explanation for beginners

Design B



Pros

- Highlights extremes and variability (temperature range + wind behavior)
- Incorporates multi-year seasonal comparison
- Adds soil temperature insights not present in Design A
- Strong for studying long-term or terrain-driven fluctuations

Cons

- More complex visuals and legends
- Does not include sunlight or humidity
- May feel denser and less introductory than Design A

Design A vs. Design B – Comparison

Design A explains the hilltop’s **overall seasonal climate cycle**, focusing on how temperature, sunlight, humidity, and moisture interact across months.

Design B highlights **extremes, variability, and multi-year wind patterns**, showing how hilltop exposure affects temperature spread, wind behavior, and soil heat.

Together, they offer two complementary perspectives:

- **Design A = broad atmospheric understanding**
- **Design B = variability, intensity, and seasonal extremes**

This satisfies the assignment requirement for two distinct design approaches for the same dashboard focus topic.

Focus Topic:

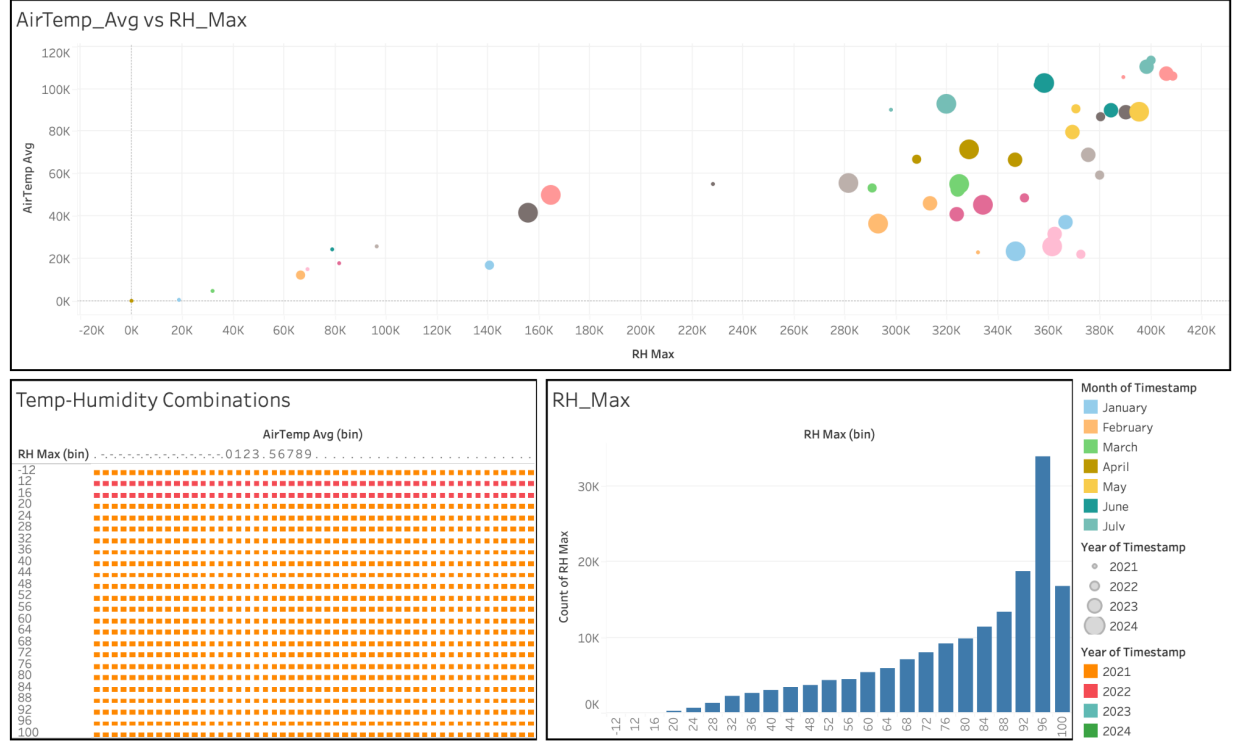
Temperature and Humidity Trends at DE Station (2021–2024)

Lead Student: Thaanvi Sudarsan Meda

Design A

Correlation and Distribution Analysis

Exploring the Relationship Between Temperature and Humidity



DISCUSSION:

Design A emphasizes **statistical relationships**. The scatterplot reveals correlation trends, the heatmap shows data density, and the histogram provides marginal distribution context. This combination moves beyond time trends to highlight physical coupling between temperature and humidity.

Pros

- Reveals correlation strength and distribution density.
- Compact, clear layout for quick pattern recognition.
- Strong quantitative comparison between variables.

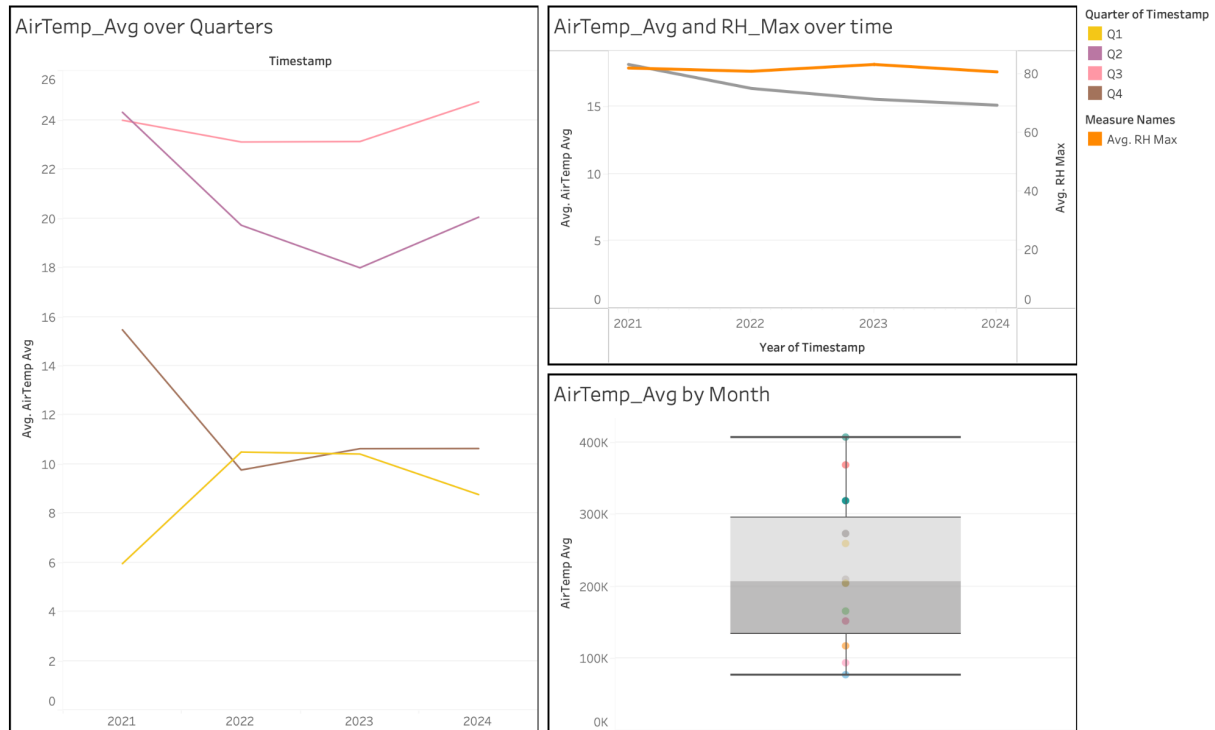
Cons

- Less temporal context—cannot track changes across specific months or days.

- Color scales must be carefully chosen to avoid misinterpretation of density.

Design B

Analysis between Temperature and Humidity
Seasonal Patterns in Temperature and Humidity



DISCUSSION:

This design emphasizes temporal continuity. The first two line charts show detailed short- and long-term patterns in temperature and humidity, while the boxplot provides a monthly summary that contextualizes extremes. Together, they help users trace how environmental conditions evolve across seasons.

Pros

- Clearly communicates seasonal and daily cycles.
- Intuitive narrative from continuous lines to monthly summary.
- Easy to detect heat waves or unusual humidity dips.

Cons

- Requires scrolling for long timelines.
- High data density may clutter the view without filters.

Design A vs. Design B – Comparison

Both designs address the same focus topic but from different analytical perspectives:

Design A = Time-oriented (temporal behavior).

Design B = Relationship-oriented (statistical correlation).

Together they provide a complete picture of Redlair’s temperature–humidity dynamics, supporting scientific understanding of local climate variability.

Focus Topic:

Seasonal Climate Patterns at the Hilltop Station for NRDE dataset (2021 – 2024)

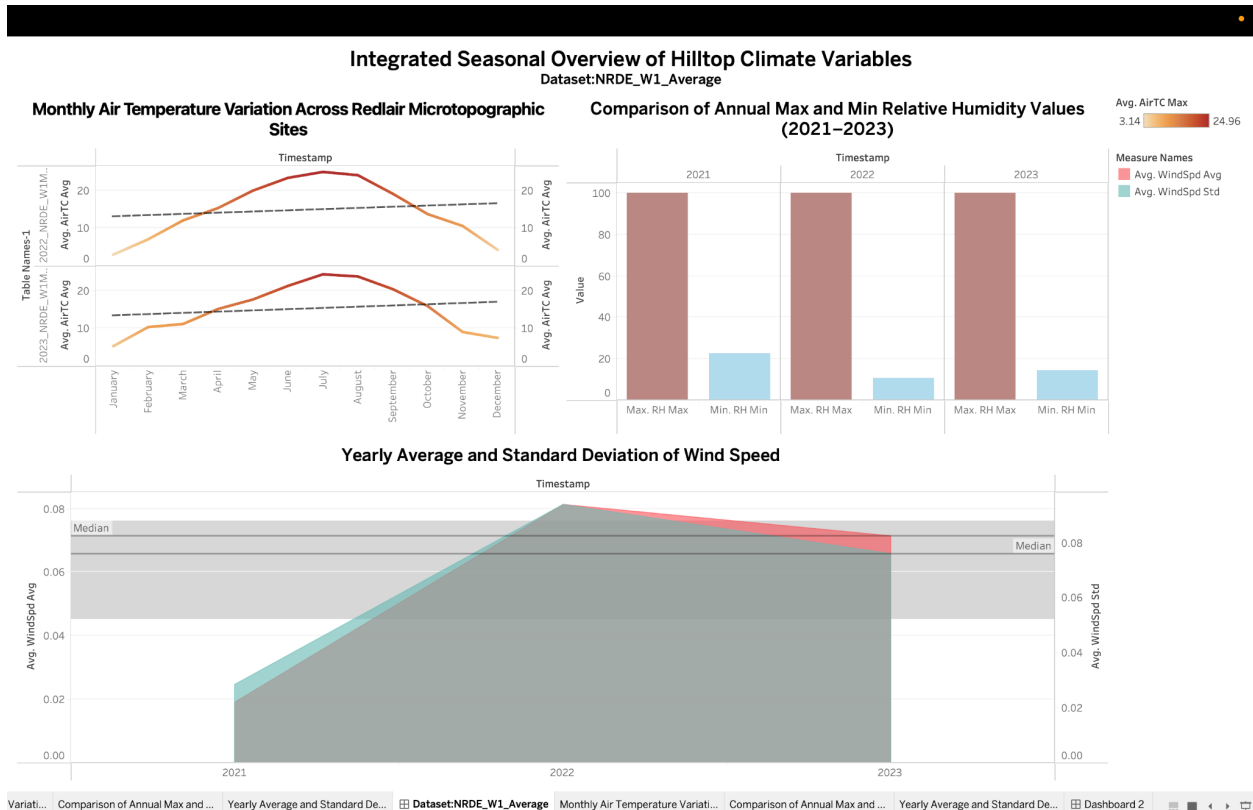
Lead Student: Ananth Pal Reddy Kandhala

Design A

Views Used:

1. **Monthly Air Temperature Variation Across Redlair Microtopographic Sites** – Line chart showing seasonal temperature changes by month.
2. **Comparison of Annual Max and Min Relative Humidity Values (2021–2023)** – Grouped bar chart comparing yearly humidity extremes.

3. Yearly Average and Standard Deviation of Wind Speed – Area chart visualizing wind speed averages and variability over time.



DISCUSSION:

Design A provides a broad climate-level overview of the hilltop microenvironment. The temperature chart reveals the monthly warming and cooling cycle, defining seasonal behavior and climate stability across microtopographic sites. The humidity comparison highlights how moisture extremes vary year-to-year, reflecting seasonal dryness and moisture recovery periods. The wind speed view adds context on atmospheric movement and exposure at the hilltop location. Together, these three views show how temperature, humidity, and wind interact to define the hilltop's annual climate signature.

Pros:

- Comprehensive representation of temperature, humidity, and wind.
- Highlights year-to-year consistency and seasonal fluctuations.
- Visually balanced and easy to interpret.
- Strong foundation for microtopography-based comparisons.

Cons:

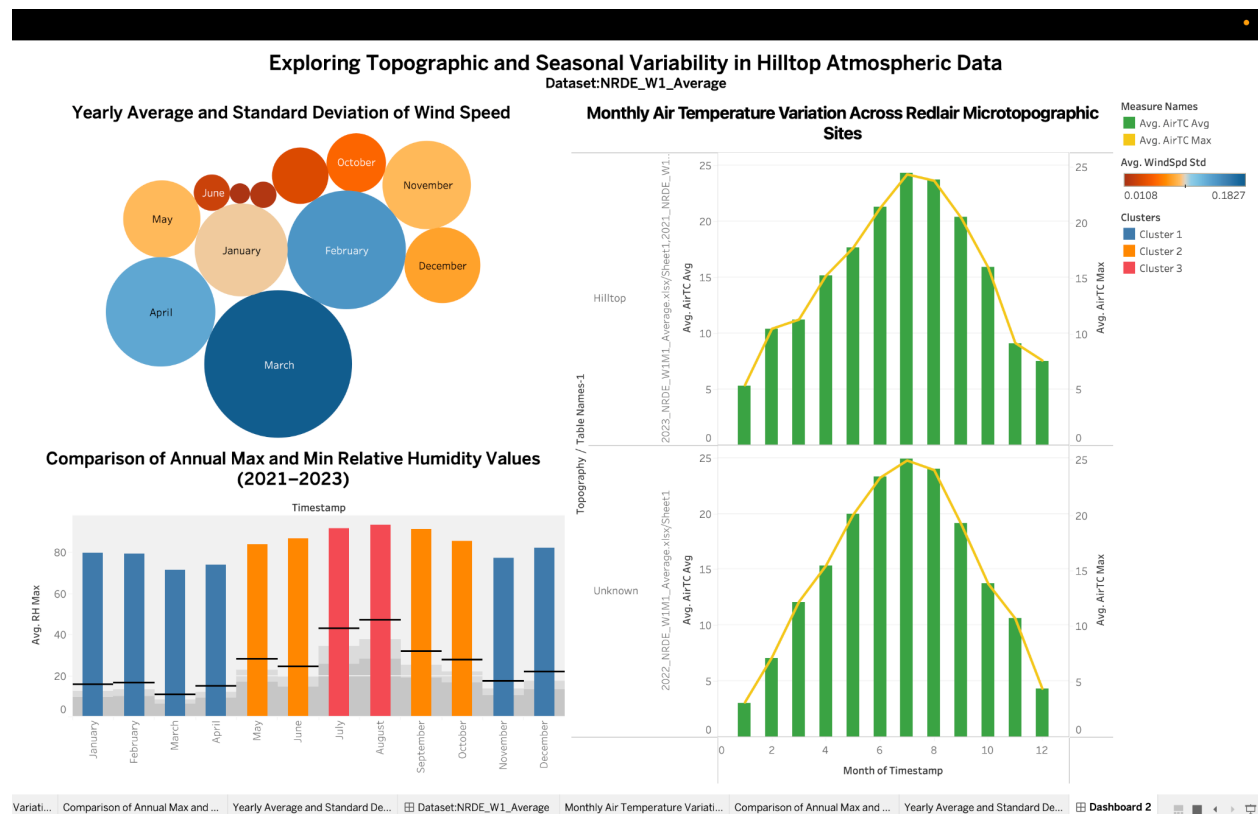
- Does not include soil or solar radiation data.

- Focuses on annual averages — less sensitive to extreme weather events.
- Wind speed variance is aggregated rather than month-specific.

Design B

Views Used:

1. Yearly Average and Standard Deviation of Wind Speed – Bubble chart showing seasonal wind speed strength and variability per month.
2. Monthly Air Temperature Variation Across Redlair Microtopographic Sites – Bar-and-line chart displaying temperature distribution by cluster and station (topography 1 vs. unknown).
3. Comparison of Annual Max and Min Relative Humidity Values (2021–2023) – Colored bar chart showing monthly humidity variations across seasons.



DISCUSSION:

Design B takes a more dynamic and comparative approach to understanding seasonal climate patterns. The bubble chart visually captures wind strength through size and color, immediately showing which months experience stronger winds (typically linked to lower humidity and temperature fluctuations). The temperature view contrasts two topographic stations, revealing how elevation influences heat retention and cooling. The humidity chart adds seasonal context

and helps compare moisture patterns throughout the year. Collectively, Design B emphasizes variability, topographic comparison, and multi-year seasonal behavior.

Pros:

- Shows seasonal wind speed variability clearly using size and color.
- Includes topographic comparisons (Hilltop vs. Unknown).
- More visually engaging and reveals inter-month patterns effectively.
- Strong for identifying seasonal extremes and month-to-month dynamics.

Cons:

- More complex and dense layout than Design A.
- Less focus on long-term trends and averages.
- Requires interpretive context for clusters and topographic terms.

Design A vs. Design B – Comparison

Design A emphasizes clarity and overall seasonal behavior, linking temperature, humidity, and wind for a macro-level view of the hilltop climate.

Design B focuses on seasonal variability and spatial comparisons, highlighting how topography and month-specific patterns influence local climate dynamics.

Together they offer complementary perspectives:

- Design A = broad climate cycle overview
- Design B = detailed seasonal and topographic variability

Focus Topic:

Seasonal Climate Patterns at the Valley Station (2021–2024)

Lead Student: Pranav Paruchuri

Design A (Dashboard 1 – Seasonal and Temperature–Humidity Trends)

DISCUSSION:

Design A provides a broad, seasonal-level analysis of the Valley microclimate, focusing on how temperature and humidity interact across the year. The monthly air-temperature line view establishes the core seasonal pattern, while the dual-axis chart of air and soil temperatures highlights how the valley’s sheltered terrain traps heat and slows cooling. The temperature–humidity relationship plot further emphasizes the valley’s high moisture retention, and the correlation view between air and soil temperature confirms the consistent thermal balance of the site. Together, these four visualizations deliver a clear narrative of how the valley maintains stable, humid conditions throughout the annual climate cycle.

Pros

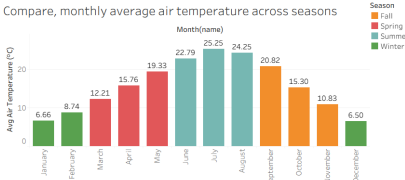
- Clearly shows **seasonal transitions** and temperature-humidity interplay.
- Demonstrates **soil–air coupling**, capturing how heat and moisture accumulate in valley terrain.
- Provides an **intuitive, story-driven overview** of annual behavior.
- Ideal baseline for comparing valley versus hilltop or midslope conditions.

Cons

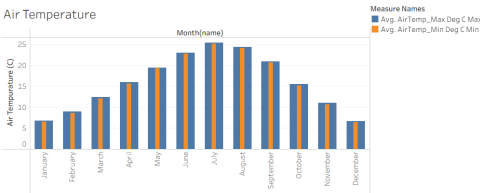
- Focuses on averages—**less detail on short-term extremes**.
- Excludes direct solar-radiation or wind-flow context.
- Some relationships (e.g., correlation scatter) may require interpretation for general audiences.

Seasonal & Temporal Trends

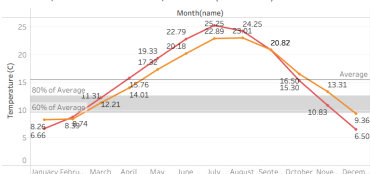
Compare, monthly average air temperature across seasons



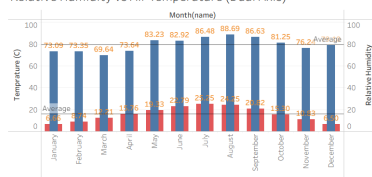
Air Temperature



Soil Temperature vs Air Temperature (Dual Line)



Relative Humidity vs Air Temperature (Dual Axis)



Design B (Dashboard 2 – Variability and Environmental Conditions)

DISCUSSION:

Design B transitions from broad seasonal trends to a closer look at **variability and environmental stability** within the valley. It analyzes daily and monthly diurnal temperature ranges for both air and soil, relative-humidity variability, and battery-voltage stability to represent environmental consistency and sensor reliability. These metrics collectively highlight the valley's reduced temperature swing, persistent humidity, and minimal operational fluctuation—evidence of its protected, low-lying microclimate.

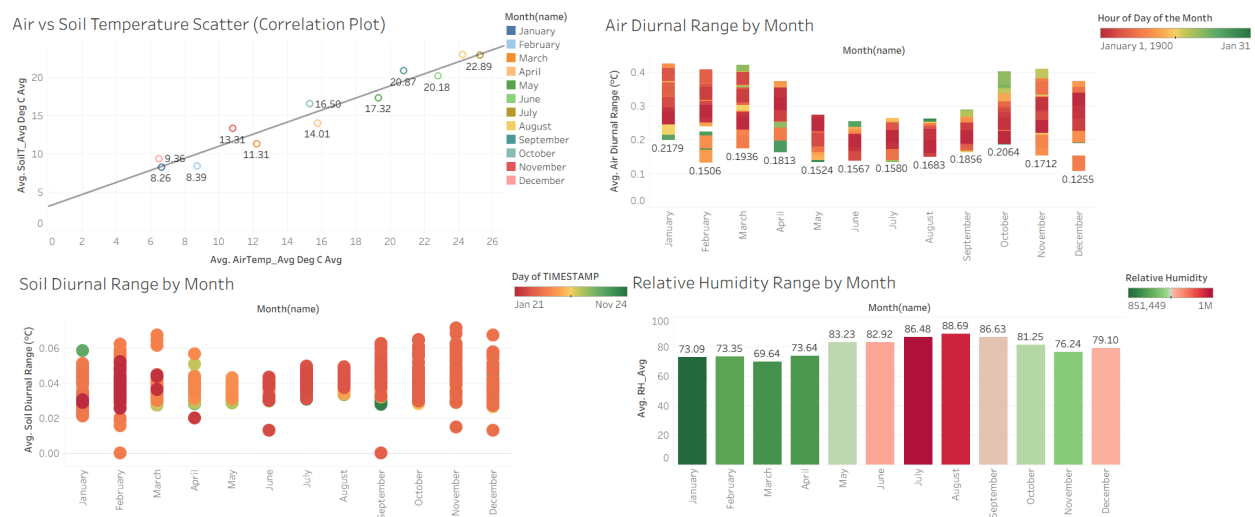
Pros

- Captures **micro-level variability** and environmental steadiness.
- Integrates **operational data** (battery health) alongside climate metrics.
- Strong for understanding **stability, persistence, and insulation** in valley ecosystems.
- Reveals subtle daily and seasonal patterns that complement Design A's broader trends.

Cons

- Less visually simple; **analytical charts dominate narrative visuals**.
- High data density may need filter controls for readability.
- Does not incorporate direct humidity–temperature relationships or sunlight data.

Variability & Environmental Conditions



Design A vs. Design B – Comparison

Aspect	Design A (Dashboard 1)	Design B (Dashboard 2)
Focus	Seasonal climate overview	Variability and stability analysis
Perspective	Broad, temporal, and comparative	Micro-scale, environmental, operational
Strengths	Clear seasonal storytelling; intuitive visuals	Reveals fine-scale consistency and sensor behavior
Limitations	Misses daily variability and operational data	Less narrative, more analytical
Complementarity	Explains <i>how the valley warms and cools</i> seasonally	Quantifies <i>how stable and humid</i> the valley remains daily

Together, Design A and Design B provide a full-spectrum understanding of the Valley station's microclimate.

- **Design A** explains the **seasonal evolution** of air, soil, and humidity conditions.
- **Design B** quantifies the **variability, stability, and operational performance** behind those trends.